

Lecture 10: Fixed Exchange Rates

The open-economy AS–AD model under a peg

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Macroeconomics B – Chapter 24

Roadmap

1. From Lecture 9 to today: what changes when the nominal exchange rate is fixed?
2. Denmark's peg as the main real-world example
3. UIP, credibility, and the loss of monetary-policy autonomy
4. The fixed-rate AS–AD model: AD, AS, and real-exchange-rate dynamics
5. Adjustment to long-run equilibrium under a peg
6. Fiscal policy under a peg: useful, but with a competitiveness hangover
7. Devaluation, expectations, and speculative pressure

Learning goals

After this lecture you should be able to:

- ▶ use UIP to explain why a credible peg pins i_t to i_t^f ,
- ▶ write the fixed-rate AS–AD model and identify the predetermined state variable,
- ▶ explain why adjustment works through inflation differentials and the real exchange rate,
- ▶ analyse a temporary fiscal expansion and trace the later loss of competitiveness,
- ▶ distinguish an unanticipated devaluation from an anticipated devaluation.

Where Lecture 9 leaves us

Lecture 9 gave the open-economy building blocks:

$$i_t = i_t^f + \Delta e_{t+1}^e$$

$$\Delta e_t^r = \Delta e_t + \pi^f - \pi_t$$

$$y_t - \bar{y} = \beta_1 e_t^r - \beta_2 (r_t - \bar{r}^f) + z_t$$

UIP

real exchange rate

open-economy demand

Today we impose one regime-specific restriction:

$$\Delta e_t = 0 \quad (\text{fixed nominal exchange rate}).$$

Central change

The nominal exchange rate no longer absorbs shocks. The real exchange rate can only move through inflation differentials.

What is a fixed exchange rate regime?

A fixed exchange rate regime sets a parity for the nominal exchange rate E .

- ▶ **Hard peg / currency board:** the central bank stands ready to buy or sell foreign currency at the parity.
- ▶ **Peg within a band:** E may fluctuate inside a narrow corridor around the parity.
- ▶ **Fixed but adjustable:** the parity can occasionally be changed through devaluation or revaluation.

The key state variable is not only the exchange rate itself, but also **credibility**: do markets believe the parity will be maintained?

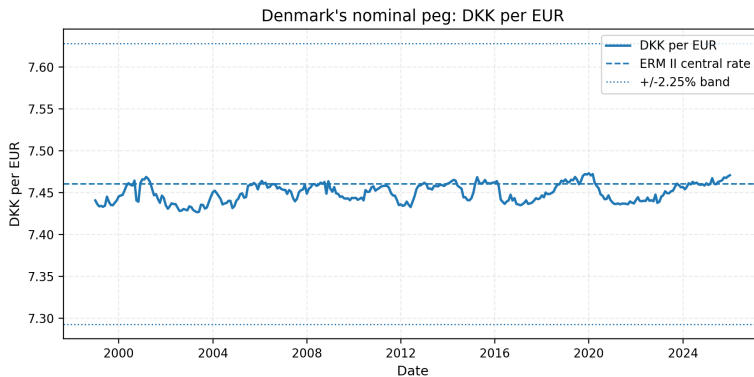
Denmark: a fixed exchange rate in practice

Denmark is the natural case study for this chapter.

- ▶ The krone is pegged to the euro within ERM II.
- ▶ The central rate is 7.46038 DKK per EUR.
- ▶ The formal narrow band is $\pm 2.25\%$, but in practice the krone is stabilised much closer to the central rate.
- ▶ Monetary policy is set to keep the krone stable against the euro, not to target Danish output directly.

Source: Danmarks Nationalbank and European Commission.

Denmark's peg in the data



The nominal exchange rate is kept close to the ERM II central rate, so nominal depreciation is not the adjustment margin.

How a central bank keeps E fixed

To prevent the nominal exchange rate from moving, the central bank can:

- ▶ buy or sell foreign currency using FX reserves,
- ▶ adjust the policy rate to affect capital flows,
- ▶ communicate its commitment to the parity.

With high capital mobility, the interest rate becomes the main defence:

weak krone pressure $\Rightarrow i_t \uparrow \Rightarrow$ capital inflows $\Rightarrow$ krone support.

Trade-off

The same rate change that supports the peg may be inappropriate for domestic stabilisation.

UIP and the trilemma

UIP says:

$$i_t = i_t^f + \Delta e_{t+1}^e.$$

Under a fully credible peg, markets expect no devaluation:

$$\Delta e_{t+1}^e = 0 \quad \Rightarrow \quad i_t = i_t^f.$$

This is the macroeconomic trilemma:

Fixed exchange rate	Free capital mobility	Independent monetary policy
Yes	Yes	No
Yes	No	Possible
No	Yes	Possible

Credible peg versus expected devaluation

If markets expect a devaluation of size δ with probability p , then

$$\Delta e_{t+1}^e = p\delta, \quad i_t = i_t^f + p\delta.$$

- ▶ A credible peg means $p\delta \approx 0$: the domestic rate can stay close to the foreign rate.
- ▶ Speculative pressure means $p\delta > 0$: the domestic rate must rise to defend the peg.
- ▶ A high interest rate can deepen the recession that made the peg doubtful in the first place.

Mechanism

A loss of credibility is contractionary before the exchange rate actually changes.

Why fix the exchange rate?

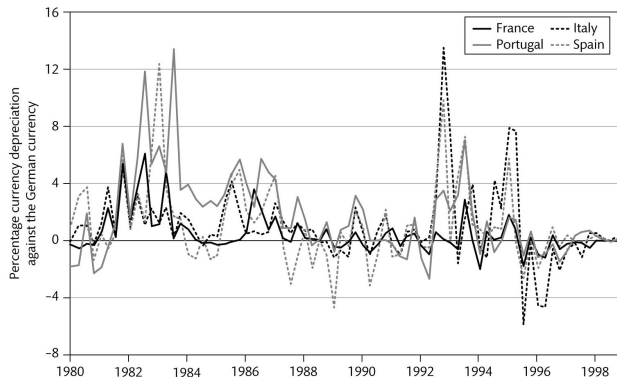
Potential benefits:

- ▶ Lower exchange-rate uncertainty for trade and investment.
- ▶ A nominal anchor: inflation expectations are tied to the anchor currency.
- ▶ Less exchange-rate volatility for firms and households with foreign exposure.

Main cost:

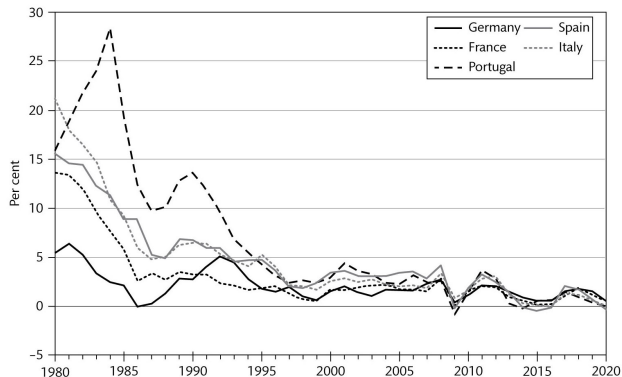
- ▶ Domestic monetary policy cannot freely stabilise output and inflation.
- ▶ Adjustment must come through wages, prices, fiscal policy, or parity changes.

Europe: exchange-rate variability before EMU



As pegs to the D-mark strengthened before EMU, exchange-rate volatility fell, lowering currency uncertainty for trade.

Europe: inflation convergence



Pegging to a low-inflation anchor coincided with lower and more aligned inflation rates.

The key state variable: the real exchange rate

The log real exchange rate is

$$e_t^r = e_t + p_t^f - p_t.$$

Taking differences gives

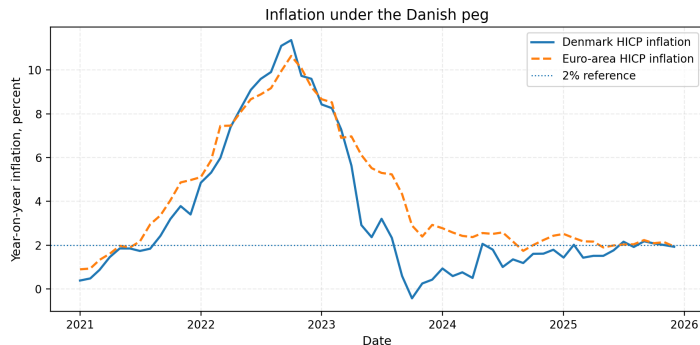
$$\Delta e_t^r = \Delta e_t + \pi^f - \pi_t.$$

Under a fixed nominal exchange rate, $\Delta e_t = 0$, so

$$\Delta e_t^r = \pi^f - \pi_t.$$

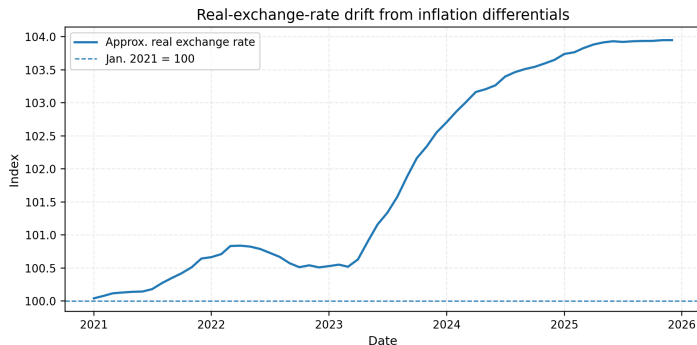
- ▶ If $\pi_t < \pi^f$: e_t^r rises, domestic goods become cheaper, competitiveness improves.
- ▶ If $\pi_t > \pi^f$: e_t^r falls, domestic goods become more expensive, competitiveness erodes.

Denmark: inflation differentials and competitiveness



Under a peg, the inflation differential relative to the euro area is the short-run source of real exchange-rate movement.

The implied real-exchange-rate drift



When Danish inflation is below euro-area inflation, the real exchange rate rises and price competitiveness improves.

Aggregate demand under a peg

Start from the open-economy demand equation:

$$y_t - \bar{y} = \beta_1 e_t^r - \beta_2 (r_t - \bar{r}^f) + \tilde{z}_t.$$

Under a credible peg:

- ▶ UIP pins $i_t = i_t^f$ and hence $r_t \approx r_t^f$.
- ▶ The real-rate term can be included in the demand shock z_t .
- ▶ Use $e_t^r = e_{t-1}^r + \pi^f - \pi_t$.

$$y_t - \bar{y} = \beta_1 e_{t-1}^r - \beta_1 (\pi_t - \pi^f) + z_t.$$

e_{t-1}^r is predetermined: it shifts AD before period- t output and inflation are determined.

Aggregate supply under a peg

With a credible low-inflation anchor, expected inflation is tied to the foreign rate:

$$\pi_t^e = \pi^f.$$

The short-run AS curve is

$$\pi_t = \pi^f + \gamma(y_t - \bar{y}) + s_t.$$

- ▶ Output above potential raises inflation above the foreign rate.
- ▶ A negative supply shock is represented by $s_t > 0$.
- ▶ In the absence of shocks, AS passes through $(\bar{y}, \pi^f)$.

The complete model

$$y_t - \bar{y} = \beta_1 e_{t-1}^r - \beta_1 (\pi_t - \pi^f) + z_t$$

AD

$$\pi_t = \pi^f + \gamma(y_t - \bar{y}) + s_t$$

AS

$$e_t^r = e_{t-1}^r + \pi^f - \pi_t$$

RER update

Timing:

1. e_{t-1}^r is inherited from the past.
2. AD and AS determine (y_t, π_t) .
3. e_t^r is updated and shifts next period's AD curve.

Long-run equilibrium: $y_t = \bar{y}$, $\pi_t = \pi^f$, and $\Delta e_t^r = 0$.

Adjustment mechanism under a peg

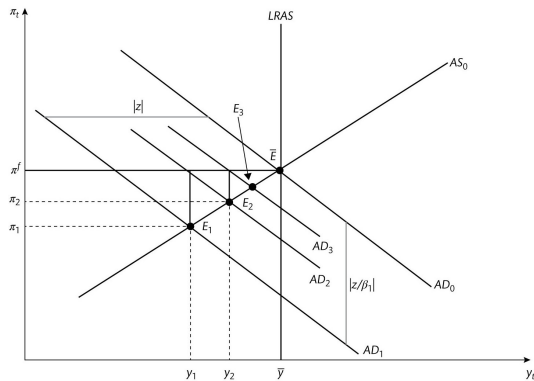
Suppose the economy starts in a recession, $y_t < \bar{y}$.

1. $y_t < \bar{y} \Rightarrow \pi_t < \pi^f$
2. $\pi_t < \pi^f \Rightarrow \Delta e_t^r > 0$
3. $e_t^r \uparrow \Rightarrow$ competitiveness improves
4. Net exports rise and next period's AD shifts right
5. y_{t+1} moves back toward $\bar{y}$

Key intuition

Under a fixed nominal exchange rate, low domestic inflation is the mechanism that produces the real depreciation needed for recovery.

Adjustment after a permanent demand shortfall



Each period of low inflation raises the real exchange rate, shifts AD right, and closes part of the output gap.

Stability in one equation

With no shocks, define $\hat{y}_t = y_t - \bar{y}$ and $\hat{\pi}_t = \pi_t - \pi^f$.

$$\hat{y}_t = \beta_1(e_{t-1}^r - \hat{\pi}_t), \quad \hat{\pi}_t = \gamma \hat{y}_t.$$

Solving gives

$$\hat{y}_t = \frac{\beta_1}{1 + \beta_1 \gamma} e_{t-1}^r, \quad e_t^r = \frac{1}{1 + \beta_1 \gamma} e_{t-1}^r.$$

Therefore the real-exchange-rate gap shrinks each period because

$$0 < \frac{1}{1 + \beta_1 \gamma} < 1.$$

Adjustment is faster when the trade channel β_1 is stronger or inflation reacts more to activity (larger γ).

Policy menu under a credible peg

Tool	Available?	Role
Monetary policy	No	tied to the foreign rate by UIP
Fiscal policy	Yes	main stabilisation instrument
Parity change	Limited	possible, but costly for credibility
Wage/price flexibility	Slow	affects real exchange-rate adjustment

Fiscal policy matters more under a peg than under an independent monetary policy regime, because the central bank cannot freely offset domestic shocks.

Temporary fiscal expansion: mechanism

A one-period increase in government spending raises z_t and shifts AD right.

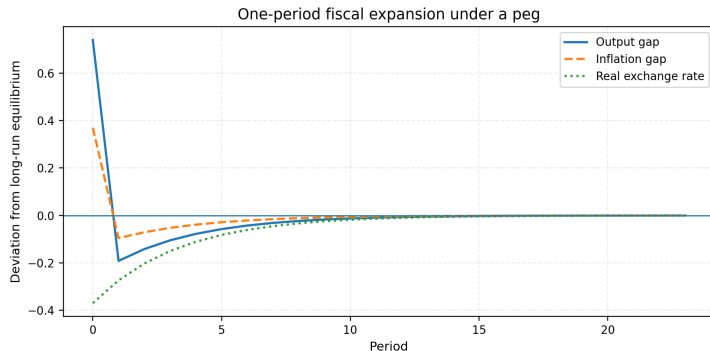
Period 1: stimulus

- ▶ $y_1 \uparrow$ and $\pi_1 \uparrow$.
- ▶ Since $\pi_1 > \pi^f$, the real exchange rate falls: competitiveness deteriorates.

Period 2: stimulus ends

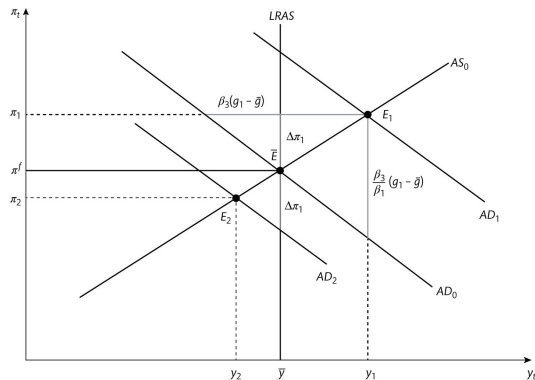
- ▶ $z_2 = 0$, but $e_1^r < e_0^r$.
- ▶ AD is now weaker because the economy has lost competitiveness.
- ▶ Output can fall below potential even without a new negative shock.

One-period fiscal expansion: simulation



The stimulus initially raises output, but the inflation spike creates a real appreciation and a later below-potential phase.

Temporary fiscal expansion in the AS–AD diagram



The fiscal impulse creates a boom on impact, but the induced inflation lowers competitiveness and shifts AD back later.

Why the post-stimulus recession appears

The RER update makes the mechanism precise:

$$e_1^r = e_0^r + (\pi^f - \pi_1).$$

Since the fiscal expansion raises inflation, $\pi_1 > \pi^f$, so

$$e_1^r < e_0^r.$$

When the stimulus ends, period-2 demand is lower because competitiveness has deteriorated:

$$y_2 - \bar{y} = \beta_1 e_1^r - \beta_1 (\pi_2 - \pi^f) + 0.$$

Lesson

Fiscal policy can stabilise demand, but under a peg it also moves future competitiveness through inflation.

Systematic countercyclical fiscal policy

Suppose the government follows a simple rule:

$$g_t - \bar{g} = -\phi(y_t - \bar{y}), \quad \phi > 0.$$

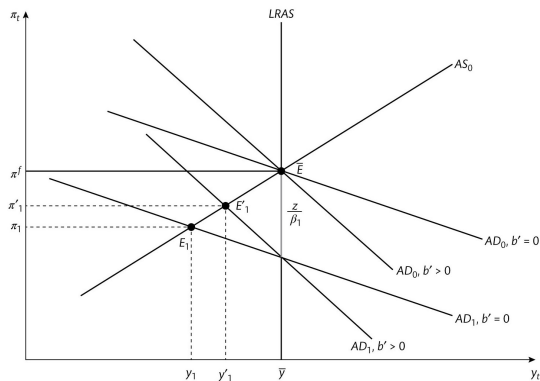
- ▶ If $y_t < \bar{y}$, spending rises automatically.
- ▶ If $y_t > \bar{y}$, spending falls automatically.

This makes the AD curve steeper:

$$\pi_t = \pi^f + e_{t-1}^r - \frac{1 + \beta_3\phi}{\beta_1}(y_t - \bar{y}) + \frac{\tilde{z}_t}{\beta_1}.$$

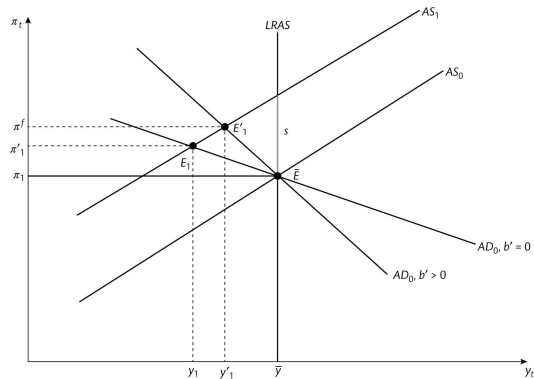
A steeper AD curve dampens demand shocks, but it can slow the later return to long-run equilibrium.

Countercyclical fiscal policy: demand shocks



Automatic stabilisation reduces the immediate fall in output and inflation after a negative demand shock.

Countercyclical fiscal policy: supply shocks



With a supply shock, stabilising output more strongly comes at the cost of a larger inflation increase.

Denmark in the recent inflation cycle

The recent Danish case illustrates the model logic.

- ▶ The exchange rate against the euro remained the nominal anchor.
- ▶ Danmarks Nationalbank followed ECB rate cuts in 2024–25 to keep the monetary-policy spread stable.
- ▶ Danish inflation was often below euro-area inflation after the 2022 energy shock.
- ▶ In the model, this means $\pi_t < \pi^f$, so the real exchange rate rises and competitiveness improves without nominal depreciation.

Connection to policy

The peg does not prevent real adjustment, but it makes adjustment operate through relative prices and demand, not through an independent Danish interest-rate path.

Parity changes: devaluation as a tool

A devaluation raises E and therefore raises e^r on impact:

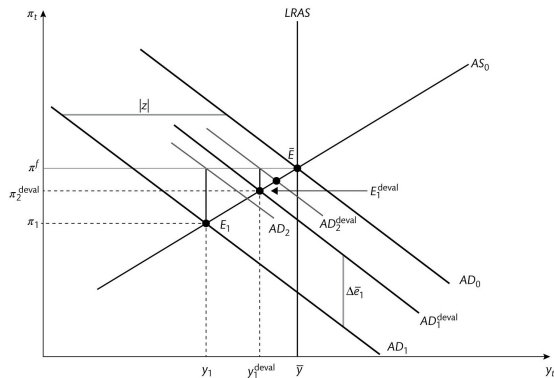
$$E \uparrow \Rightarrow e^r \uparrow \Rightarrow NX \uparrow \Rightarrow AD \text{ shifts right.}$$

But the timing of expectations is decisive:

- ▶ **Unanticipated devaluation:** no pre-devaluation rate spike.
- ▶ **Anticipated devaluation:** UIP raises the domestic rate before the parity changes.

Devaluation can restore competitiveness, but it may weaken the credibility of the peg going forward.

Unanticipated devaluation



A surprise devaluation improves competitiveness immediately and shifts AD outward; later inflation erodes part of the gain.

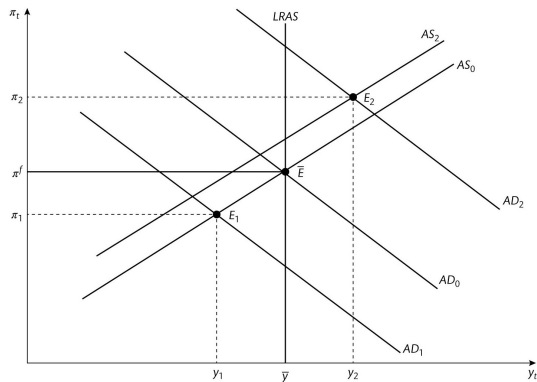
Anticipated devaluation: the UIP channel

If markets expect devaluation, $\Delta e_{t+1}^e > 0$, so UIP requires

$$i_t = i_t^f + \Delta e_{t+1}^e > i_t^f.$$

- ▶ To defend the peg today, the central bank must accept higher rates.
- ▶ Higher rates lower private demand before the devaluation occurs.
- ▶ The economy can enter recession while waiting for the parity change.

Anticipated devaluation in the AS-AD diagram



Expected devaluation raises the domestic interest rate, shifts demand left before the parity change, and weakens the policy tool.

Speculative pressure under a peg

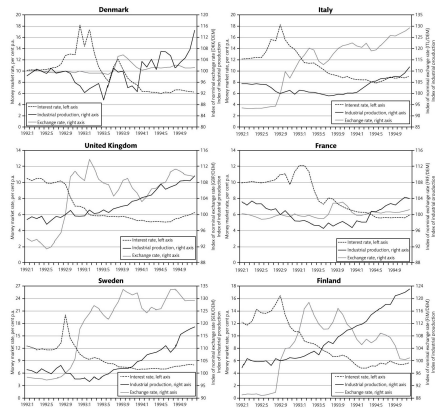
A fragile peg can generate a self-reinforcing loop:

recession $\Rightarrow$ doubts about parity $\Rightarrow \Delta e^e \uparrow \Rightarrow i \uparrow \Rightarrow$ deeper recession.

This is why credibility is central:

- ▶ A credible peg allows the domestic rate to stay close to the foreign rate.
- ▶ A doubtful peg requires a risk/devaluation premium in the domestic rate.
- ▶ Previous devaluations may make future devaluations more likely in expectations.

European stress episodes



Defending pegs during stress episodes often required high interest rates, which deepened recessions and sometimes preceded devaluation.

Active exercise: simulate a fiscal expansion

Work in pairs for 15–20 minutes. Use the notebook or write the equations directly.

Parameters:

$$\beta_1 = 0.7, \quad \gamma = 0.5, \quad \pi^f = 0.02.$$

Initial state: $e_0^r = 0$, $\hat{y}_0 = 0$. Shocks:

$$z_1 = 1, \quad z_t = 0 \text{ for } t \geq 2, \quad s_t = 0.$$

1. Iterate the model for 20 periods.
2. Plot $\hat{y}_t$, $\hat{\pi}_t$, and e_t^r .
3. Identify the period where the post-stimulus recession starts.
4. Change β_1 from 0.4 to 1.0. What happens to the recovery speed?

Summary

- ▶ A credible peg with free capital mobility implies $i_t = i_t^f$; monetary policy is not an independent stabilisation tool.
- ▶ Under a fixed nominal exchange rate, real adjustment happens through inflation differentials: $\Delta e_t^r = \pi^f - \pi_t$.
- ▶ A recession lowers inflation, raises competitiveness, shifts AD right, and gradually closes the output gap.
- ▶ Fiscal policy is useful under a peg, but temporary stimulus can create a competitiveness hangover.
- ▶ Devaluation works best when unanticipated, but the act of devaluing can damage future credibility.
- ▶ Next lecture: floating exchange rates and how the adjustment mechanism changes.

Appendix A: Deriving AD under a peg

Start from the preliminary open-economy AD relation:

$$y_t - \bar{y} = \beta_1 e_t^r - \beta_2 (r_t - \bar{r}^f) + \tilde{z}_t.$$

A credible peg implies $i_t = i_t^f$ and $r_t \approx r_t^f$, so define

$$z_t \equiv \tilde{z}_t - \beta_2 (r_t^f - \bar{r}^f).$$

With $\Delta e_t = 0$,

$$e_t^r = e_{t-1}^r + \pi^f - \pi_t.$$

Substitution gives

$$y_t - \bar{y} = \beta_1 (e_{t-1}^r + \pi^f - \pi_t) + z_t = \beta_1 e_{t-1}^r - \beta_1 (\pi_t - \pi^f) + z_t.$$

Appendix B: Within-period solution

Define $\hat{y}_t = y_t - \bar{y}$ and $\hat{\pi}_t = \pi_t - \pi^f$:

$$\begin{aligned}\hat{y}_t &= \beta_1 e_{t-1}^r - \beta_1 \hat{\pi}_t + z_t, \\ \hat{\pi}_t &= \gamma \hat{y}_t + s_t.\end{aligned}$$

Substitute AS into AD:

$$\hat{y}_t = \beta_1 e_{t-1}^r - \beta_1 (\gamma \hat{y}_t + s_t) + z_t.$$

Collect terms:

$$(1 + \beta_1 \gamma) \hat{y}_t = \beta_1 e_{t-1}^r + z_t - \beta_1 s_t.$$

Hence

$$\boxed{\hat{y}_t = \frac{\beta_1 e_{t-1}^r + z_t - \beta_1 s_t}{1 + \beta_1 \gamma}}, \quad \boxed{\hat{\pi}_t = \frac{\gamma \beta_1 e_{t-1}^r + \gamma z_t + s_t}{1 + \beta_1 \gamma}}.$$

Appendix C: Stability without shocks

With $z_t = s_t = 0$:

$$\hat{y}_t = \frac{\beta_1}{1 + \beta_1 \gamma} e_{t-1}^r, \quad \hat{\pi}_t = \frac{\gamma \beta_1}{1 + \beta_1 \gamma} e_{t-1}^r.$$

The real exchange rate evolves as

$$e_t^r = e_{t-1}^r - \hat{\pi}_t = e_{t-1}^r - \frac{\gamma \beta_1}{1 + \beta_1 \gamma} e_{t-1}^r = \frac{1}{1 + \beta_1 \gamma} e_{t-1}^r.$$

Since $0 < 1/(1 + \beta_1 \gamma) < 1$, the real-exchange-rate gap converges to zero. The output gap inherits the same root:

$$\hat{y}_{t+1} = \frac{1}{1 + \beta_1 \gamma} \hat{y}_t.$$

Appendix D: Temporary fiscal expansion

Let $z_1 > 0$, $z_t = 0$ for $t \geq 2$, $s_t = 0$, and $e_0^r = 0$.

Period 1:

$$\hat{y}_1 = \frac{z_1}{1 + \beta_1 \gamma} > 0, \quad \hat{\pi}_1 = \frac{\gamma z_1}{1 + \beta_1 \gamma} > 0.$$

RER update:

$$e_1^r = 0 - \hat{\pi}_1 = -\frac{\gamma z_1}{1 + \beta_1 \gamma} < 0.$$

Period 2:

$$\hat{y}_2 = \frac{\beta_1 e_1^r}{1 + \beta_1 \gamma} = -\frac{\beta_1 \gamma}{(1 + \beta_1 \gamma)^2} z_1 < 0.$$

Thus a positive temporary demand shock can generate a later recession through lost competitiveness.

Appendix E: Countercyclical fiscal rule

Suppose the demand shock includes government spending:

$$z_t = \beta_3(g_t - \bar{g}) + \tilde{z}_t, \quad g_t - \bar{g} = -\phi \hat{y}_t.$$

Then

$$\hat{y}_t = \beta_1(e_{t-1}^r - \hat{\pi}_t) - \beta_3\phi \hat{y}_t + \tilde{z}_t.$$

So

$$(1 + \beta_3\phi)\hat{y}_t = \beta_1(e_{t-1}^r - \hat{\pi}_t) + \tilde{z}_t.$$

Solving for $\hat{\pi}_t$ gives

$$\hat{\pi}_t = e_{t-1}^r - \frac{1 + \beta_3\phi}{\beta_1}\hat{y}_t + \frac{\tilde{z}_t}{\beta_1}.$$

The AD curve is steeper when ϕ is larger, so demand shocks are dampened more strongly.

Appendix F: Anticipated devaluation

Suppose markets assign probability p to a devaluation of size δ next period:

$$\Delta e_{t+1}^e = p\delta.$$

UIP implies

$$i_t = i_t^f + p\delta.$$

If expected inflation remains anchored at π^f , then the real rate rises as well:

$$r_t = i_t - \pi_{t+1}^e = i_t^f + p\delta - \pi^f.$$

The IS channel is contractionary:

$$r_t \uparrow \Rightarrow C_t \downarrow, I_t \downarrow \Rightarrow y_t \downarrow.$$

This is the anticipation effect: the expectation of devaluation reduces output before the devaluation occurs.